Eduardo José Barroso, Ph.D.

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Born: 11 June 1995 (Age: 31)

Research Experience

08/2024 — Present
Postdoctoral Researcher, LAPP, Annecy, France.
Advisors: Thibault Guillemin, Dominique Boutigny.

10/2022 — 07/2023
Ph.D Exchange Researcher, Laboratoire de Physique Subatomique et de Cosmologie (LPSC), Grenoble, France.
Advisor: Céline Combet.

2020 — 2024
Ph.D Researcher, Universidade Estadual de Londrina (UEL), Londrina, Brazil.
Thesis: Large scale structure formation: from primordial black holes in bouncing cosmology to galaxy clusters abundance.
Advisor: Sandro Dias Pinto Vitenti.

Education

2020 – 2024
Ph.D in Physics, Universidade Estadual de Londrina (UEL), Londrina, Brazil.
Thesis: Large scale structure formation: from primordial black holes in bouncing cosmology to galaxy clusters abundance.
Advisor: Sandro Dias Pinto Vitenti. Defended on 11 July 2024.

10/2022 — 07/2023
Ph.D Exchange Program, Laboratoire de Physique Subatomique et de Cosmologie (LPSC), Grenoble, France.
Advisor: Céline Combet. Funded by CAPES (process 88887.510837/2020-00).

08/2018 — 07/2020
Masters in Physics, Universidade Estadual de Londrina (UEL), Londrina, Brazil.
Dissertation: Bosonization approach for the quantum Hall effect.
Advisor: Paula Fernanda Bienzobaz. Co-advisor: Pedro Rogerio Sergi Gomes.

2013-2018
Bachelor in Chemical Engineering, Universidade Estadual de Maringá (UEM), Maringá, Brazil.

Collaboration Memberships

- LSST DESC Full member (since 2023)
- LSST DESC Builder (since 2026), a lifetime status for major contributions to DESC software and community, with authorship on all DESC Key Papers

Research Publications

Journal Articles

1. X. Ye, L. F. Demétrio, **E. J. Barroso**, S.-F. Yan, and N. Pinto-Neto, "Primordial black hole formation in dust-radiation bouncing cosmologies," *Journal of Cosmology and Astroparticle Physics*, vol. 2026, no. 07, p. 083, 2026. DOI: 10.1088/1475-7516/2026/07/083.

2. **E. J. Barroso**, L. F. Demétrio, S. D. P. Vitenti, and X. Ye, "Primordial black hole formation in a dust bouncing model," *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 01, p. 052, 2025. DOI: 10.1088/1475-7516/2025/01/052.

3. C. Payerne, Z. Zhang, M. Aguena, C. Combet, T. Guillemin, M. Ricci, N. Amouroux, C. Avestruz, **E. J. Barroso**, A. Farahi, E. Kovacs, C. Murray, M. M. Rau, E. S. Rykoff, and S. J. Schmidt, "Weak lensing mass-richness relation of redMaPPer clusters in LSST DESC DC2 simulations," *Astronomy and Astrophysics*, vol. 700, A34, 2025. DOI: 10.1051/0004-6361/202554107.

4. S. D. P. Vitenti, F. de Simoni, M. Penna-Lima, and **E. J. Barroso**, "AutoKnots: Adaptive knot allocation for spline interpolation," *Astronomy and Computing*, vol. 53, 100970, 2025. DOI: 10.1016/j.ascom.2025.100970.

5. S. D. P. Vitenti and **E. J. Barroso**, "APES: Approximate posterior ensemble sampler," *Monthly Notices of the Royal Astronomical Society*, vol. 525, no. 1, pp. 73–90, 2023. DOI: 10.1093/mnras/stad2245.

In LSST DESC Internal Review

1. **E. J. Barroso**, C. Payerne, M. Aguena, C. Combet, S. Elles, T. Guillemin, M. Ricci, S. D. P. Vitenti, J. Zuntz, N. Amouroux, C. Avestruz, D. Boutigny, N. E. Chisari, N. Lorenzo, E. Pedersen, Y. Zhang, and LSST Dark Energy Science Collaboration, "Preparation for an end-to-end LSST DESC pipeline for cosmology with galaxy clusters," In LSST DESC internal review, 2026.

2. M. Aguena, C. Combet, S. Fu, C. Payerne, M. Ricci, C. Avestruz, **E. J. Barroso**, C. L. de Oliveira, H. Fan, A. Niemiec, T. Shin, R. Solomon, R. Srinivasan, Y. Zhang, C. Zhou, and LSST Dark Energy Science Collaboration, "Getting ready for analysing Rubin data: CLMM v2, the updated LSST-DESC cluster weak lensing mass modeling library for cosmology," In LSST DESC internal review, 2026.

3. R. Solomon, M. Aguena, C. Benoist, T. Guillemin, **E. J. Barroso**, D. Boutigny, L. da Costa, N. Amouroux, and S. J. Schmidt, "Characterizing the WaZP cluster finder on LSST simulated images," In LSST DESC internal review, 2026.

Oral Talks

2026
- Online Seminar at APC, Paris, France, 08/10/2026. Title: Cluster Cosmology with LSST-DESC
- Talk at COSMO-26, International Conference on Particle Physics and Cosmology, Leiden, Netherlands. Title: From Simulations to Early Data: Galaxy Cluster Cosmology with LSST-DESC

2025
- LSST France meeting talk, LPSC, Grenoble, France, 25/11/2025. Title: Introduction to DESC Tools
- Talk at the Enigmass+ General Assembly, LAPP, Annecy, France, 07/11/2025. Title: Galaxy Cluster Cosmology with LSST
- Seminar at LAPP, Annecy, France. Title: Cosmology with sky surveys
- Online Seminar at Observatório Nacional, Rio de Janeiro, Brasil, 07/07/2025. Title: Galaxy Cluster Cosmology with LSST
- LSST France meeting talk, Orsay, France, 12/06/2025. Title: DESC Cluster Pipeline Updates: Full Chain Analysis

2024
- LSST France meeting talk, APC, Paris, France, 11/2024. Title: Advancements in the DESC Cluster Pipeline: Current State and Cosmological Analysis with Firecrown

2023
- LSST France meeting talk, LPSC, Grenoble, France, 06/2023. Title: APES: Approximate Posterior Ensemble Sampler

Workshops

2025
- Organized the Cluster Pipeline workshop at LAPP, Annecy, France.

Supervision

2025 **Supervision of L3 internship**, Jules Maquinghen "Fitting the scaling relation of Wazp DC2 from LSST"(LAPP, France)

Teaching

2022 **Teaching Assisting plus Monitor** for Quantum Mechanics and Data Analysis at Universidade Estadual de Londrina UEL $\sim 60h$

2021 **Teaching Assisting plus Monitor** for Statistical Mechanics at Universidade Estadual de Londrina UEL $\sim 60h$

2014 **Calculus Monitor** at Universidade Estadual de Maringa UEM $\sim 60h$

Exam Committees

2026 **Ph.D qualifying exam committee**, Henrique Cardoso Naves Lettieri, "Cosmology-dependent covariance in galaxy cluster number counts: consequences for parameter inference", Universidade Estadual de Londrina, Londrina, Brasil, 27/07/2026.

2025 **Master's defense committee**, Thais Mikami Ornellas, "Relações concentração-massa em escalas de grupos e aglomerados de galáxias: impacto nas restrições de massa e de parâmetros cosmológicos", Universidade Estadual de Londrina, Londrina, Brasil, 29/08/2025.

Master's qualifying exam committee, Thais Mikami Ornellas, "Relações concentração-massa em escalas de grupos e aglomerados de galáxias: impacto nas restrições de massa e de parâmetros cosmológicos", Universidade Estadual de Londrina, Londrina, Brasil, 16/05/2025.

Outreach

2025 Facilitator of the "Rubin First Look" event at LPSC, Grenoble, France

Participation on the "Rubin First Look" event at LAPP, Annecy, France

Skills

Languages — Portuguese (native), English (fluent), French (advanced), German (basic).

Coding — Python, C, LaTeX, MCMC, Data Analysis.

Software — SQL, Git.

Misc. — Academic research, teaching, numerical cosmology.